

Next Generation Sequencing (NGS)

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Tracking Information

Publication Number

100-3

Manual Section Number

90.2

Manual Section Title

Next Generation Sequencing (NGS)

Version Number

2

Effective Date of this Version

01/27/2020

Implementation Date

11/13/2020

Description Information

Benefit Category

Diagnostic Laboratory Tests

Diagnostic Services in Outpatient Hospital

Diagnostic Tests (other)

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

A. General

Clinical laboratory diagnostic tests can include tests that, for example, predict the risk associated with one or more genetic variations. In addition, in vitro companion diagnostic laboratory tests provide a report of test results of genetic variations and are essential for the safe and effective use of a corresponding therapeutic product. Next Generation Sequencing (NGS) is one technique that can measure one or more genetic variations as a laboratory diagnostic test, such as when used as a companion in vitro diagnostic test.

This National Coverage Determination (NCD) is only applicable to diagnostic lab tests using NGS for somatic (acquired) and germline (inherited) cancer. Medicare Administrative Contractors (MACs) may determine coverage of diagnostic lab tests using NGS for RNA sequencing and protein analysis.

Indications and Limitations of Coverage

B. Nationally Covered Indications

1. Somatic (Acquired) Cancer

Effective for services performed on or after March 16, 2018, the Centers for Medicare & Medicaid Services (CMS) has determined that Next Generation Sequencing (NGS) as a diagnostic laboratory test is reasonable and necessary and covered nationally, when performed in a Clinical Laboratory Improvement Amendments (CLIA)-certified laboratory, when ordered by a treating physician, and when all of the following requirements are met:

- a. Patient has:
 - i. either recurrent, relapsed, refractory, metastatic, or advanced stage III or IV cancer; and
 - ii. not been previously tested with the same test using NGS for the same cancer genetic content, and
 - iii. decided to seek further cancer treatment (e.g., therapeutic chemotherapy).
- b. The diagnostic laboratory test using NGS must have:
 - i. Food & Drug Administration (FDA) approval or clearance as a companion in vitro diagnostic; and,
 - ii. an FDA-approved or -cleared indication for use in that patient's cancer; and,
 - iii. results provided to the treating physician for management of the patient using a report template to specify treatment options.

2. Germline (Inherited) Cancer

Effective for services performed on or after January 27, 2020, CMS has determined that NGS as a diagnostic laboratory test is reasonable and necessary and covered nationally for patients with germline (inherited) cancer, when performed in a CLIA-certified laboratory, when ordered by a treating physician and when all of the following requirements are met:

- a. Patient has:
 - i. ovarian or breast cancer; and,
 - ii. a clinical indication for germline (inherited) testing for hereditary breast or ovarian cancer; and,
 - iii. a risk factor for germline (inherited) breast or ovarian cancer; and
 - iv. not been previously tested with the same germline test using NGS for the same germline genetic content.
- b. The diagnostic laboratory test using NGS must have all of the following:
 - i. FDA-approval or clearance; and,
 - ii. results provided to the treating physician for management of the patient using a report template to specify treatment options.

C. Nationally Non-Covered Indications

1. Somatic (Acquired) Cancer

Effective for services performed on or after March 16, 2018, NGS as a diagnostic laboratory test for patients with acquired (somatic) cancer are non-covered if the cancer patient does not meet the criteria noted in section B.1., above.

D. Other

1. Somatic (Acquired) Cancer

Effective for services performed on or after March 16, 2018, Medicare Administrative Contractors (MACs) may determine coverage of NGS as a diagnostic laboratory test for patients with advanced cancer only when the test is performed in a CLIA-certified laboratory, when ordered by a treating physician, and when the patient has:

- a. either recurrent, relapsed, refractory, metastatic, or advanced stages III or IV cancer; and,
- b. not been previously tested with the same test using NGS for the same cancer genetic content, and
- c. decided to seek further cancer treatment (e.g., therapeutic chemotherapy).

2. Germline (Inherited) Cancer

Effective for services performed on or after January 27, 2020, MACs may determine coverage of NGS as a diagnostic laboratory test for patients with germline (inherited) cancer only when the test is performed in a CLIA-certified laboratory, when ordered by a treating physician, when results are provided to the treating physician for management of the patient and when the patient has:

- a. any cancer diagnosis; and,
- b. a clinical indication for germline (inherited) testing of hereditary cancers; and,
- c. a risk factor for germline (inherited) cancer; and,
- d. not been previously tested with the same germline test using NGS for the same germline genetic content.

(This NCD last reviewed January 2020)

Transmittal Information

Transmittal Number

10346

Coverage Transmittal Link

<https://www.cms.gov/files/document/R10346NCD.pdf> 

Revision History

04/2026 - Transmittal 13623 issued February 05, 2026, is being rescinded and replaced by Transmittal 13760, dated April 30, 2026, to remove business requirement 14356.4 and the corresponding spreadsheet for NCD 250.4 Treatment of Actinic Keratosis as this NCD was moved to the October 2026 CR. All other information remains the same. ([TN 13760](#)) (CR14356)

02/2026 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determinations (NCDs). No policy is being changed as a result of these updates. ([TN 13623](#)) (CR14356)

09/2025 - Transmittal 13400 issued September 05, 2025, is being rescinded and replaced by Transmittal 13438, dated September 30, 2025, to add a new attachment to ensure the functionality of the NCD zip URL is working properly and to update the Policy section. All other information remains the same.
The purpose of this Change Request (CR) is to provide a maintenance update of ICD-10 coding changes specific to NCDs. ([TN 13438](#)) (CR14197)

08/2025 - The purpose of this Change Request (CR) is to provide a maintenance update of ICD-10 coding changes specific to NCDs. ([TN 13375](#)) (CR14197)

03/2025 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determinations (NCDs). No policy is being changed as a result of these updates. ([TN 13097](#)) (CR13939)

05/2024 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determinations (NCDs). No policy is being changed as a result of these updates. ([TN 12626](#)) (CR13596)

01/2024 - Transmittal 12355 issued November 09, 2023, is being rescinded and replaced by Transmittal 12444, January 4, 2024 to add clarifying verbiage to NCD 90.2, business requirement 13278.2, and related edits to NCD 90.2, Next Generation Sequencing, attached spreadsheet. All other information remains the same. ([TN 12444](#)) (CR13278)

01/2024 - Transmittal 12350 issued November 03, 2023, is being rescinded and replaced by Transmittal 12440 dated January 3, 2024, to make changes to NCD 90.2, Next Generation Sequencing, spreadsheet to align with revisions being made to CR 13278. All other information remains the same. ([TN 12440](#)) (CR13391)

11/2023 - The purpose of this Change Request (CR) is to provide a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 12355](#)) (CR13278)

11/2023 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National

Coverage Determinations (NCDs). No policy is being changed as a result of these updates. ([TN 12350](#)) (CR13391)

10/2023 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determinations (NCDs). No policy is being changed as a result of these updates. ([TN 12319](#)) (CR13391)

08/2023 - The purpose of this Change Request (CR) is to provide a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. ([TN 12184](#)) (CR13278)

05/2023 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determination (NCDs). No policy is being changed as a result of these updates. ([TN 12017](#)) (CR13166)

06/2022 - Transmittal 11055, dated October 21, 2021, is being rescinded and replaced by Transmittal 11461, dated, June 21, 2022 to revise BR 12483.1 and corresponding spreadsheet to align with changes made in previous CR 12124 and later CR 12705. All other information remains the same. ([TN 11461](#)) (CR12483)

06/2022 - Transmittal 11400, dated May 4, 2022, is being rescinded and replaced by Transmittal 11460, dated, June 17, 2022, to update NCD 90.2, NGS, spreadsheet to conform with changes in CR 12124, and change the implementation date for all business requirements except 12705.6 to 30 days from issuance of this correction. All other information remains the same. ([TN 11460](#)) (CR12705)

06/2022 - Transmittal 10832, dated June 2, 2021, is being rescinded and replaced by Transmittal 11453, dated, June 10, 2022, to revise NCD 90.2, NGS, revises business requirement 12124.2 and 12124.2.1 and its associated spreadsheet of coding by retaining all ICD-10 NOC diagnosis codes proposed for deletion effective July 1, 2022. ([TN 11453](#)) (CR12124)

05/2022 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 11400](#)) (CR12705)

10/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 11055](#)) (CR12483)

06/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 10832](#)) (CR12124)

05/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the

result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 10804](#)) (CR12124)

03/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 10624](#)) (CR12124)

09/2020 - The purpose of this change request is to inform contractors that effective for dates of service on and after January 27, 2020, CMS has determined that NGS, as a diagnostic laboratory test, is reasonable and necessary and covered nationally for patients with germline (inherited) cancer when performed in a CLIA-certified laboratory, when ordered by a treating physician and when specific requirements are met. ([TN 10346](#)) (CR11837)

07/2020 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 10261](#)) (CR11905)

06/2020 - Transmittal 2439, dated February 21, 2020, is being rescinded and replaced by Transmittal 10193, dated, June 19, 2020 remove Current Procedural Technology (CPT) code 0048U from business requirement 11655.1 and corresponding removals of CPT 0048U and its associated diagnosis codes from the National Coverage Determination (NCD) 90.2 Next Generation Sequencing (NGS) spreadsheet. This revision is necessary because the CPT code does not meet the policy criteria in NCD 90.2 for NGS. All other information remains the same. ([TN 10193](#)) (CR11655)

05/2020 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html>, along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-related changes to NCDs continue to be implemented via the current, longstanding NCD process. ([TN 10092](#)) (CR11749)

02/2020 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html>, along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-

related changes to NCDs continue to be implemented via the current, longstanding NCD process. ([TN 2439](#)) (CR11655)

04/2019 - Transmittal 214, dated March 6, 2019, is being rescinded and replaced by Transmittal 215, dated, April 10, 2019 to incorporate missing CPT codes on the diagnosis code attachments. All other information remains the same. ([TN 215](#)) (CR10878)

03/2019 - Transmittal 210, dated November 30, 2018, is being rescinded and replaced by Transmittal 214, dated, March 6, 2019 to extend the implementation date 30 days and update the attached diagnosis code list. This instruction is being re-communicated to include attachment that was omitted. The Transmittal number, Dated issued and all other information remain the same. ([TN 214](#)) (CR10878)

11/2018 - The purpose of this Change Request (CR) is to inform contractors that effective March 16, 2018, the Centers for Medicare & Medicaid Services covers diagnostic laboratory tests using next generation sequencing when performed in a Clinical Laboratory Improvement Amendments- certified laboratory when ordered by a treating physician and when specific requirements are met.

THIS CHANGE REQUEST (CR) AND PUBLICATION (PUB.) 100-03 MANUAL TRANSMITTAL REFLECTS THE AGENCY'S FINAL DECISION DATED MARCH 16, 2018, REGARDING THE NATIONAL COVERAGE DETERMINATION (NCD) 90.2, ON NEXT GENERATION SEQUENCING (NGS). A SUBSEQUENT CR WILL BE RELEASED AT A LATER DATE THAT CONTAINS A PUB. 100-04 CLAIMS PROCESSING MANUAL UPDATE AND FURTHER, ACCOMPANYING INSTRUCTIONS. UNTIL THAT TIME, THE MEDICARE ADMINISTRATIVE CONTRACTORS (MACS) SHALL BE RESPONSIBLE FOR IMPLEMENTING NCD 90.2. ([TN 210](#)) (CR10878)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Next Generation Sequencing \(NGS\) for Medicare Beneficiaries with Advanced Cancer \(CAG-00450N\)](#)
- [First reconsideration for Next Generation Sequencing \(NGS\) for Medicare Beneficiaries with Advanced Cancer \(CAG-00450R\)](#)

Additional Information

[Other Versions](#)

Title	Version	Effective Between	
Next Generation Sequencing (NGS)	2	01/27/2020 - N/A	You are here
Next Generation Sequencing (NGS)	1	03/16/2018 - 01/27/2020	View